06.10.20

K-map:

Karnaugh-map:

Binary: (2 bit change)

00 → 0
01 → 1
10 → 2
11 → 3

K-map: (only one bit change)

00 → 0
01 → 1
11 → 3
10 → 2

- **Use of k-map:** Simplification

(*) $F = \bar{A}\bar{B}\bar{C} + \bar{B}C + \bar{A}B$

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	$\bar{A}\bar{B}\bar{C}$ 000	$\bar{A}\bar{B}C$ 001	$\bar{A}BC$ 011	$\bar{A}B\bar{C}$ 010
A 1	$A\bar{B}\bar{C}$ 100	$A\bar{B}C$ 101	ABC 111	$AB\bar{C}$ 110

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	$\bar{A}\bar{B}\bar{C}$ 000 — 0 (1)	$\bar{A}\bar{B}C$ 001 — 1 (1)	$\bar{A}BC$ 011 — 3 1	$\bar{A}B\bar{C}$ 010 — 2 1
A 1	$A\bar{B}\bar{C}$ 100 — 4 0	$A\bar{B}C$ 101 — 5 (1)	ABC 111 — 7 0	$AB\bar{C}$ 110 — 6 0

Normally:

$$F = \bar{A}B\bar{C} + \bar{B}C + \bar{A}B$$

$$= \bar{A} + \bar{B} + \bar{C} + \bar{B}C + \bar{A}B$$

$$= \bar{A}(1+B) + \bar{B}(1+C) + \bar{C}.$$

$$= \bar{A}\cdot 1 + \bar{B}\cdot 1 + \bar{C}$$

$$= \bar{A} + \bar{B} + \bar{C}$$

$$F = \bar{A}B\bar{C} + \bar{B}C + \bar{A}B$$

$$= \bar{B}(\bar{C} + \bar{A}\bar{C}) + \bar{A}B$$

$$= \bar{B}(C + \bar{A}) + \bar{A}B$$

$$= \bar{B}C + \bar{A}\bar{B} + \bar{A}B$$

$$= \bar{B}C + \bar{A}(B + \bar{B})$$

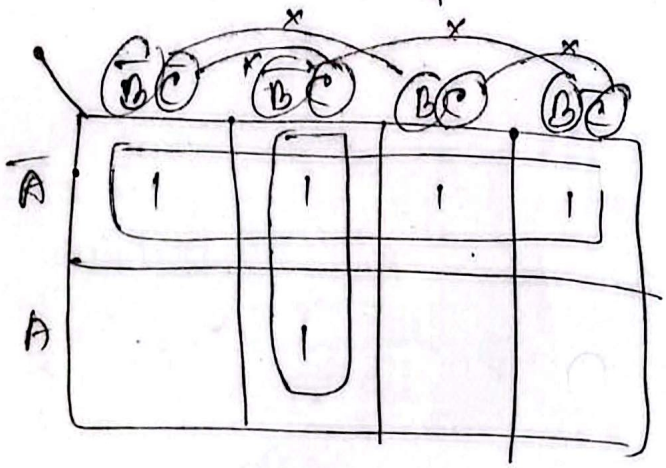
$$= \bar{A} + \bar{B}C$$

K-map:

Allowed:



• B ও $\bar{B}$ দুই ঘরে গ্রুপ করা,



$$\therefore F = \bar{A}B\bar{C} + \bar{B}C + \bar{A}B$$

$$= \bar{A} + \bar{B}C$$

(*) Using k-map, simplify the expression: $Y = \bar{A}\bar{B}\bar{C} + \bar{B}C + \bar{A}B$.

⇒

- K-map
- Final answer.

HW:

(*). $Y = \bar{C}(A\bar{B}\bar{D} + D) + A\bar{B}C + \bar{D}$,

. Simplify them using k-map.